Calorimetry lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use an energy balance to analyze mixing and heat transfer.

SUCCESS CRITERIA - I can...

1. Identify hot and cold samples and their masses.
2. Balance energy gained and lost in a simplified calculation.
3. Discuss heat exchange with the container and surroundings.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “calorimetry” and “isolated” to explain your answer. Sentence starters are optional.

calorimetry: using temperature changes to track energy transfers

isolated: exchanging no relevant quantity with surroundings in this model

heat: energy transferred because of a temperature difference

01 Warm-up: bring back an earlier idea

A **3 kg** sample has c=**500 J**/(kg·°C) and warms by **7°C** without changing phase. **Find Q**.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $Q_{hot} + Q_{cold} = 0$; $Q = mc(T_f - T_i)$.
Ideal isolated mixing, no phase change. Heat lost is negative; heat gained is positive. Real surroundings may exchange energy.

Worked reasoning: Two equal masses of the same liquid at **20°C** and **40°C** mix in an insulated container. The final **30°C** makes the hot sample lose the same energy the cool sample gains.

Your turn: Draw heat-transfer arrows and **show the two equal temperature** changes.

Sentence starter: In my work, I used … to show … . I labeled it … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Equal masses of the same liquid at **20°C** and **60°C** are mixed in an ideal insulated container. **Predict the final temperature**. **Explain how you reached your** answer.

Word help: justify = **support your answer** with a reason or evidence

Sentence starter: I predict … . I used … to make this prediction because … .

20°C

60°C

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

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Continue down each column.

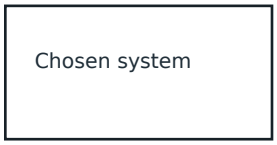
Sentence starters are optional. You may plan in any language.

04 Apply the physics

In a mixing trial, the warm water loses **300 J**. First assume only the cold water receives that energy. Then consider a cup that absorbs **50 J**. **Draw both accounts** and **find the cold water's energy** gain in each case.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

Plan a table for masses, initial temperatures, final temperature and a heat-loss observation. Why should the final temperature be measured promptly after mixing?

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A spring stores **4 J** at a given compression. Within its Hooke's-law range, find the energy at **6** times that compression.
Mark equilibrium, displacement and restoring-force direction.

Word help: compression = shortening or squeezing relative to an unstressed length

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

07 Exit check

An ideal mixture of **30°C** and **70°C** samples reaches **80°C** with no energy added. Is that possible? Explain.

Word help: ideal = using stated simplifying conditions

Sentence starter: I think … . I can point to …, which shows …, so … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Mix one cup of **20°C** water with two equal-size cups of **50°C** water, neglecting losses. **Predict whether** final temperature is above or below **35°C**, then **calculate it**.

Sentence starter: I predict … . I used … to make this prediction because … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3